

Mini Enterprise Network Infrastructure

Capstone Project Report – Network Security & VAPT Assessment

Online Cybersecurity Internship Project

Prepared By

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Role

Network Security Specialist / VAPT Trainee

Assessment Type

Vulnerability Assessment, Traffic Analysis & System Hardening

Assessment Environment

Virtual Laboratory Subnet (10.162.238.0/24)

Submission Date

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Document Control

Field	Details
Document Title	Design, Implementation, and Security Assessment of a Mini Enterprise Network
Project Type	Capstone Project – Online Internship
Prepared By	Naveen William
Role	Network Security Specialist / VAPT Trainee
Target Infrastructure	Heterogeneous Enterprise Network (Windows 10 & Ubuntu Server)
Core Technology	Nmap, Wireshark, UFW, PowerShell, Docker, DVWA, Linux Subsystem
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Revision History

Version	Date	Description	Author
1.0	August 2026	Initial Network VAPT & Hardening Report	Naveen William
1.1	August 2026	Added Windows 10 Host Hardening & Screenshot Mapping	Naveen William

2. Executive Summary

Enterprise subnets require balanced network architecture, service isolation, and continuous vulnerability assessment to mitigate exposure to malicious actors. This technical report details the end-to-end design, virtual deployment, security assessment, traffic analysis, and cross-platform system hardening of a mini enterprise network environment.

The infrastructure consists of a three-host virtual enterprise topology model within the **10.162.238.0/24** subnet: an Attack/Investigator Host (**10.162.238.51**), an Employee Workstation running Windows 10 (**10.162.238.20**), and an Enterprise Web/Database Server hosting containerized web applications (DVWA) alongside legacy network daemons (**vsftpd**). Port scanning and active enumeration using Nmap identified exposed legacy services, including unencrypted FTP on TCP port 21, as well as SMB and RPC exposure on the workstation. Traffic analysis conducted via Wireshark captured cleartext credentials transmitted over unencrypted protocol streams.

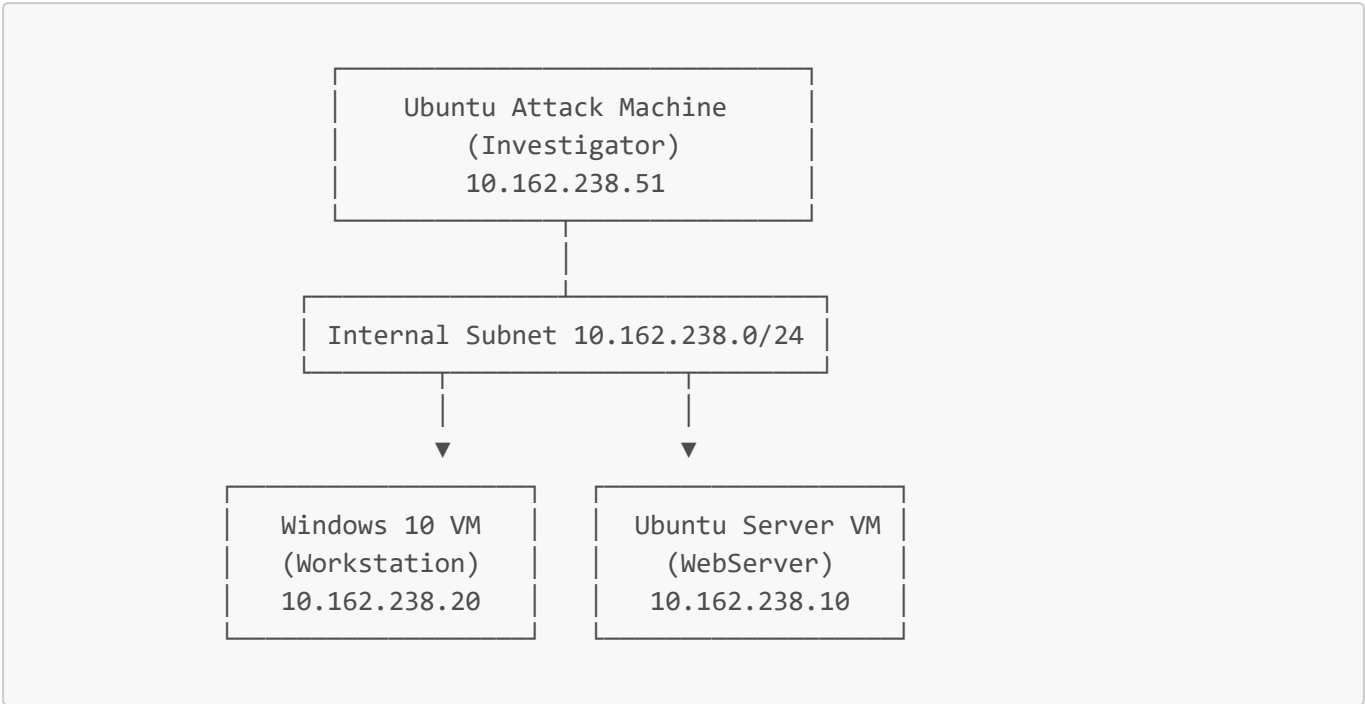
Following the initial discovery phase, cross-platform hardening measures were executed: service deprecation and stateful Uncomplicated Firewall (UFW) policy enforcement on the server endpoint, alongside ICMP dropping, SMB firewall rules, and mandatory SMB security signing on the Windows endpoint. Post-remediation verification scans confirmed that unencrypted entry vectors were blocked (**filtered**) and host visibility was restricted while preserving core operational services.

3. Scope & Objectives

- **Network Architecture & IP Planning:** Design and document a structured IP addressing scheme connecting isolated virtual endpoints within a single broadcast domain.
- **Virtual Deployment & Interconnectivity:** Configure and validate bidirectional ICMP reachability across Windows and Linux operating systems.
- **Reconnaissance & Service Enumeration:** Execute network sweeps and deep service scans using Nmap to map open ports, operational daemons, hostnames, and OS details.
- **Network Traffic Analysis:** Capture active protocol streams with Wireshark to demonstrate the security risks of cleartext authentication protocols.
- **Cross-Platform System Hardening:** Remediate identified exposure vectors by terminating legacy daemons on Linux (UFW) and hardening SMB/ICMP policies on Windows 10.
- **Validation Scanning:** Perform post-remediation Nmap scans to verify the effectiveness of the implemented remediation controls.

4. System Architecture & IP Addressing

The environment simulates a small enterprise network layout consisting of an administration/audit machine, a client workstation endpoint, and a multi-service web and database server.



Device Name	Operating System	Subnet IP Address	MAC Address	Role / Exposed Services
Attacker Host	Ubuntu Linux	10.162.238.51	10:6f:d9:bb:19:07	Reconnaissance, Traffic Analysis, Audit
Workstation	Windows 10 Pro	10.162.238.20	08:00:27:ad:fd:ba	Employee Endpoint (MSRPC, NetBIOS, SMB)
Enterprise Server	Ubuntu Server	10.162.238.10	Target Interface	Web App (DVWA: 80), FTP (vsftpd: 21)

5. Reconnaissance & Service Enumeration

Active network enumeration was conducted from the primary attack host (10.162.238.51) against all target endpoints inside the 10.162.238.0/24 subnet.

5.1 Subnet Host Discovery

```
nmap -sn 10.162.238.0/24
```

Result: Confirmed active host ICMP response across 10.162.238.20 (Windows 10) and the Ubuntu Server host prior to firewall hardening.

5.2 Workstation Enumeration (10.162.238.20)

A target port scan was executed against standard administrative and file-sharing ports:

```
nmap -sV -sC -p 135,139,445,3389 -oN win10_init_scan.txt 10.162.238.20
```

Target Output Log (win10_init_scan.txt):

```
Nmap scan report for 10.162.238.20
Host is up (0.00044s latency).

PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?
3389/tcp   closed ms-wbt-server
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
|_ nbstat: NetBIOS name: DESKTOP-HQS02MS, NetBIOS user: <unknown>, NetBIOS MAC:
08:00:27:ad:fd:ba (Oracle VirtualBox virtual NIC)
|_ smb2-time:
|   date: 2026-08-25T10:36:01
|_ start_date: N/A
|_ smb2-security-mode:
|   3:1:1:
|_ Message signing enabled but not required
```

- Findings: Windows RPC (135), NetBIOS (139), and SMB (445) are active. Remote Desktop (3389) is currently disabled (closed). SMB message signing was enabled but not required, meaning SMB sessions were not configured to require signing. This configuration can increase exposure to certain relay and man-in-the-middle attack scenarios.

5.3 Server Endpoint Enumeration (Initial Scan)

Deep enumeration was executed against the Ubuntu Server VM to identify active application services:

```
nmap -sV -sC -O -p- 10.162.238.10 -oN server_init_scan.txt
```

Target Output Log ([server_init_scan.txt](#)):

```
# Nmap 7.94SVN scan initiated Tue Aug 25 16:03:52 2026 as: nmap -sV -sC -O -p- -oN
server_init_scan.txt 10.162.238.10
Nmap scan report for 10.162.238.10
Host is up (0.00038s latency).
Not shown: 65532 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.5
22/tcp    open  ssh      OpenSSH 10.2p1 Ubuntu 2ubuntu3.5 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http     Apache httpd 2.4.25 ((Debian))
|_http-server-header: Apache/2.4.25 (Debian)
| http-title: Login :: Damn Vulnerable Web Application (DVWA) v1.10 *Develop...
|_Requested resource was login.php
| http-robots.txt: 1 disallowed entry
|_/
| http-cookie-flags:
|   /:
|     PHPSESSID:
|_     httponly flag not set
MAC Address: 08:00:27:D0:45:C3 (Oracle VirtualBox virtual NIC)
No exact OS matches for host (If you know what OS is running on it, see
https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.94SVN%E=4%D=8/25%OT=21%CT=1%CU=30436%PV=Y%DS=1%DC=D%G=Y%M=08002
OS:7%TM=6A8D6FA7%P=x86_64-pc-linux-gnu)SEQ(SP=FA%GCD=1%ISR=10C%TI=Z%CI=Z%TS
OS:=20)SEQ(SP=FA%GCD=1%ISR=10C%TI=Z%CI=Z%II=I%TS=20)OPS(O1=M5B4ST11NW9%O2=M
OS:5B4ST11NW9%O3=M5B4NNT11NW9%O4=M5B4ST11NW9%O5=M5B4ST11NW9%O6=M5B4ST11)WIN
OS:(W1=FE88%W2=FE88%W3=FE88%W4=FE88%W5=FE88%W6=FE88)ECN(R=Y%DF=Y%T=40%W=FAF
OS:0%O=M5B4NNSNW9%CC=Y%Q=)T1(R=Y%DF=Y%T=40%S=0%A=S+%F=AS%RD=0%Q=)T2(R=N)T3(
OS:R=N)T4(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T5(R=Y%DF=Y%T=40%W=0%S=Z
OS:%A=S+%F=AR%O=%RD=0%Q=)T6(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T7(R=Y
OS:%DF=Y%T=40%W=0%S=Z%A=S+%F=AR%O=%RD=0%Q=)U1(R=Y%DF=N%T=40%IPL=164%UN=0%RI
OS:PL=G%RID=G%RIPCK=G%RUCK=G%RUD=G)IE(R=Y%DFI=N%T=40%CD=S)

Network Distance: 1 hop
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at
https://nmap.org/submit/ .
# Nmap done at Tue Aug 25 16:04:15 2026 -- 1 IP address (1 host up) scanned in
23.10 seconds
```

- Findings:
 - TCP Port 21 (FTP): Running vsftpd service allowing cleartext network access.
 - TCP Port 80 (HTTP): Exposing Docker-hosted Damn Vulnerable Web Application (DVWA) server environment.

6. Traffic Analysis & Vulnerability Proof

To evaluate protocol security across the subnet, traffic captures were recorded into network_log.pcapng using Wireshark on interface wlp3s0.



6.1 Cleartext FTP Credential Interception

During an active FTP session initiated toward the server endpoint, authentication packets were sniffed and reconstructed using Wireshark's Follow TCP Stream feature.

- **Protocol:** File Transfer Protocol (FTP)
- **Observed Packets:** Unencrypted payload headers
- **Captured Credentials:** USER analyst / PASS root

Attribute	Vulnerability Assessment Details
Finding	Unencrypted File Transfer Protocol (FTP) Authentication Stream
Protocol / Port	FTP / TCP Port 21
Authentication Type	Cleartext Payload Transmission
Captured Evidence	images/plaintext_ftp_cred.png (USER analyst / PASS root)
Impact	Credentials can be observed, intercepted, and reused by any local network attacker executing passive sniffing or ARP poisoning attacks.
Remediation	Deprecate FTP daemon (vsftpd); enforce secure encrypted transfers using SFTP / SSH (TCP Port 22).

The image displays a Wireshark packet capture window and a terminal window. The Wireshark window is titled "Capturing from wlp3s0" and shows a list of 180 packets. The selected packet is packet 137, which is an FTP response (220) from 10.162.238.51 to 10.162.238.10. The terminal window shows the command prompt of an analyst@elk-server: ~, where the user has executed the command `ftp analyst@10.162.238.10`. The terminal output shows the FTP session details, including the connection to 10.162.238.10, the response 220 (vsFTPD 3.0.5), the request for a password, the successful login response 230, the remote system type (UNIX), and the use of binary mode for file transfers.

Wireshark packet capture window showing the following details:

No.	Time	Source	Destination	Protocol	Length	Info
137	10.207503899	10.162.238.10	10.162.238.51	FTP	86	Response: 220 (vsFTPD 3.0.5)
139	10.207640998	10.162.238.51	10.162.238.10	FTP	80	Request: USER analyst
141	10.207858796	10.162.238.10	10.162.238.51	FTP	100	Response: 331 Please specify the password.
171	12.529053346	10.162.238.51	10.162.238.10	FTP	77	Request: PASS root
172	12.562448384	10.162.238.10	10.162.238.51	FTP	89	Response: 230 Login successful.
174	12.562679682	10.162.238.51	10.162.238.10	FTP	72	Request: SYST
175	12.563029971	10.162.238.10	10.162.238.51	FTP	85	Response: 215 UNIX Type: L8
176	12.563193984	10.162.238.51	10.162.238.10	FTP	72	Request: FEAT
177	12.563459481	10.162.238.10	10.162.238.51	FTP	81	Response: 211-Features:
178	12.563507418	10.162.238.10	10.162.238.51	FTP	87	Response: EPRT
180	12.563639868	10.162.238.10	10.162.238.51	FTP	110	Response: PASV

Terminal window showing the command prompt and the output of the FTP command:

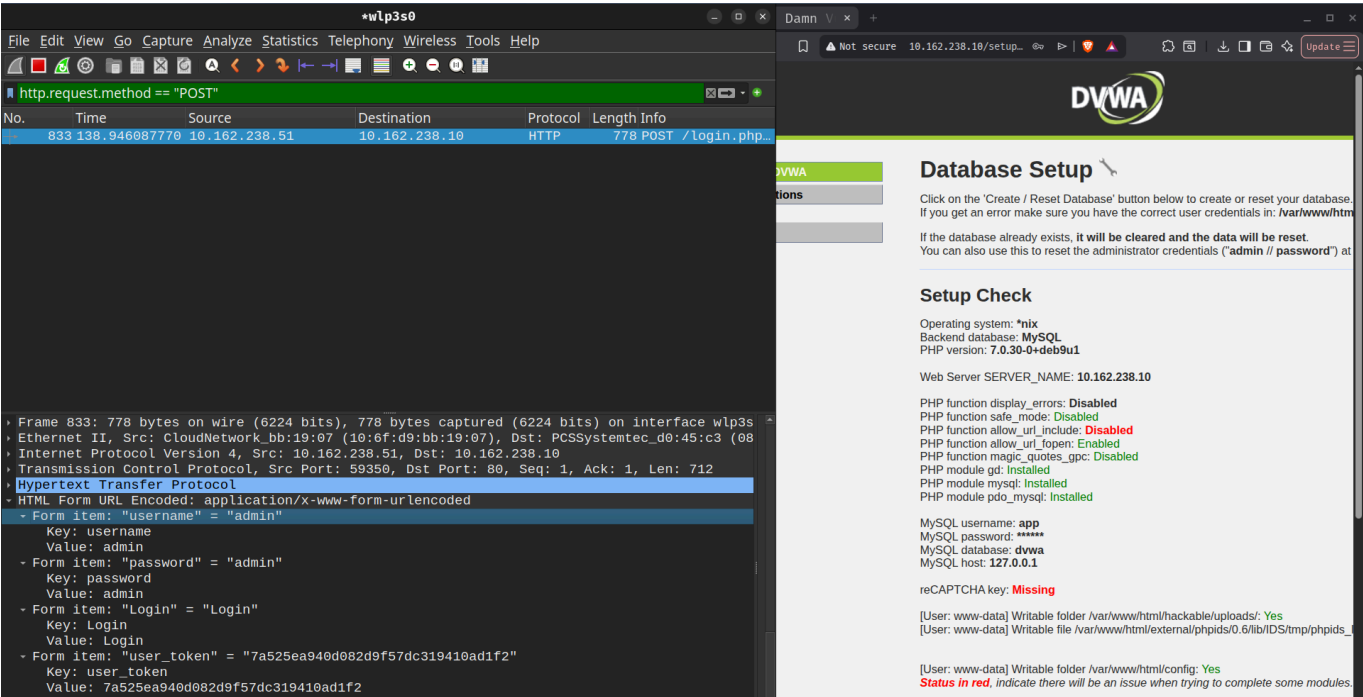
```
analyst@elk-server: ~  
> ftp analyst@10.162.238.10  
Connected to 10.162.238.10.  
220 (vsFTPD 3.0.5)  
331 Please specify the password.  
Password:  
230 Login successful.  
Remote system type is UNIX.  
Using binary mode to transfer files.  
ftp>
```

6.2 Cleartext HTTP Web Application Capture

HTTP POST requests sent to the DVWA web application were captured on TCP port 80, revealing raw parameter fields passed across the wire.

- **Risk Assessment:** Transmitting credentials over cleartext channels exposes enterprise accounts to eavesdropping, ARP spoofing, and session hijacking vectors across local segments.

Attribute	Vulnerability Assessment Details
Finding	Unencrypted HTTP POST Request Transmitting Web Authentication Data
Protocol / Port	HTTP / TCP Port 80
Authentication Type	Cleartext HTTP Form Data
Captured Evidence	images/plaintext_dvwa_cred.png (username=admin, password=password)
Impact	Session tokens, administrative credentials, and sensitive POST data are completely exposed to local network eavesdropping and session hijacking.
Remediation	Enforce Transport Layer Security (TLS 1.3) with HTTPS (TCP Port 443) and redirect all cleartext HTTP requests.



7. System Hardening & Mitigation

Defensive hardening measures were executed across both Linux server and Windows client endpoints to shrink the total attack surface.

7.1 Server Endpoint Hardening (Ubuntu Linux)

- **Legacy Service Termination:** The unencrypted FTP daemon was stopped and permanently disabled from system startup:

```
sudo systemctl stop vsftpd
```

```
analyst@elk-server:~$ sudo systemctl status vsftpd
* vsftpd.service - vsftpd FTP server
   Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-08-25 10:22:22 UTC; 44min ago
   Invocation: 82be368a23cb443792c1f4c64d2c154f
   Main PID: 4161 (vsftpd)
     Tasks: 1 (limit: 2155)
    Memory: 1M (peak: 41.9M)
       CPU: 605ms
   CGroup: /system.slice/vsftpd.service
           └─4161 /usr/sbin/vsftpd /etc/vsftpd.conf

Aug 25 10:34:11 elk-server vsftpd[5233]: pam_unix(vsftpd:auth): check pass; user unknown
Aug 25 10:34:11 elk-server vsftpd[5233]: pam_unix(vsftpd:auth): authentication failure; logname= uid=0 euid=0 tty=ftp ruser=anonymous rhost=::ffff:10.162.238.51
Aug 25 10:34:11 elk-server vsftpd[5237]: pam_unix(vsftpd:auth): check pass; user unknown
Aug 25 10:34:11 elk-server vsftpd[5237]: pam_unix(vsftpd:auth): authentication failure; logname= uid=0 euid=0 tty=ftp ruser=anonymous rhost=::ffff:10.162.238.51
Aug 25 10:38:32 elk-server vsftpd[5293]: pam_unix(vsftpd:auth): check pass; user unknown
Aug 25 10:38:32 elk-server vsftpd[5293]: pam_unix(vsftpd:auth): authentication failure; logname= uid=0 euid=0 tty=ftp ruser=admin rhost=::ffff:10.162.238.51
Aug 25 10:39:30 elk-server vsftpd[5301]: pam_unix(vsftpd:auth): check pass; user unknown
Aug 25 10:39:30 elk-server vsftpd[5301]: pam_unix(vsftpd:auth): authentication failure; logname= uid=0 euid=0 tty=ftp ruser=admin rhost=::ffff:10.162.238.51
Aug 25 10:40:30 elk-server unix_chkpwd[5326]: password check failed for user (analyst)
Aug 25 10:40:30 elk-server vsftpd[5323]: pam_unix(vsftpd:auth): authentication failure; logname= uid=0 euid=0 tty=ftp ruser=analyst rhost=::ffff:10.162.238.51
analyst@elk-server:~$ sudo systemctl stop vsftpd
analyst@elk-server:~$ sudo systemctl status vsftpd
* vsftpd.service - vsftpd FTP server
   Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; enabled; preset: enabled)
   Active: inactive (dead) since Tue 2026-08-25 11:06:32 UTC; 3s ago
   Duration: 44min 9.420s
   Invocation: 82be368a23cb443792c1f4c64d2c154f
   Process: 4161 ExecStart=/usr/sbin/vsftpd /etc/vsftpd.conf (code=killed, signal=TERM)
   Main PID: 4161 (code=killed, signal=TERM)
    Mem peak: 41.9M
       CPU: 605ms

Aug 25 10:34:11 elk-server vsftpd[5237]: pam_unix(vsftpd:auth): authentication failure; logname= uid=0 euid=0 tty=ftp ruser=anonymous rhost=::ffff:10.162.238.51
Aug 25 10:38:32 elk-server vsftpd[5293]: pam_unix(vsftpd:auth): check pass; user unknown
Aug 25 10:38:32 elk-server vsftpd[5293]: pam_unix(vsftpd:auth): authentication failure; logname= uid=0 euid=0 tty=ftp ruser=admin rhost=::ffff:10.162.238.51
Aug 25 10:39:30 elk-server vsftpd[5301]: pam_unix(vsftpd:auth): check pass; user unknown
Aug 25 10:39:30 elk-server vsftpd[5301]: pam_unix(vsftpd:auth): authentication failure; logname= uid=0 euid=0 tty=ftp ruser=admin rhost=::ffff:10.162.238.51
Aug 25 10:40:30 elk-server unix_chkpwd[5326]: password check failed for user (analyst)
Aug 25 10:40:30 elk-server vsftpd[5323]: pam_unix(vsftpd:auth): authentication failure; logname= uid=0 euid=0 tty=ftp ruser=analyst rhost=::ffff:10.162.238.51
Aug 25 11:06:32 elk-server systemd[1]: Stopping vsftpd.service - vsftpd FTP server...
Aug 25 11:06:32 elk-server systemd[1]: vsftpd.service: Deactivated successfully.
Aug 25 11:06:32 elk-server systemd[1]: Stopped vsftpd.service - vsftpd FTP server.
analyst@elk-server:~$ _
```

- **UFW Firewall Configuration:** A strict default-deny firewall posture was established using Uncomplicated Firewall (UFW):

```
sudo ufw default deny incoming
sudo ufw default allow outgoing
sudo ufw allow 80/tcp      # HTTP - Web Access
sudo ufw allow 22/tcp      # SSH - Encrypted Admin
sudo ufw enable
```

```
analyst@elk-server:~$ sudo ufw status
Status: inactive
analyst@elk-server:~$ sudo ufw enable
Firewall is active and enabled on system startup
analyst@elk-server:~$ sudo ufw default deny incoming
Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)
analyst@elk-server:~$ sudo ufw default allow outgoing
Default outgoing policy changed to 'allow'
(be sure to update your rules accordingly)
analyst@elk-server:~$ sudo ufw allow 80/tcp
Rule added
Rule added (v6)
analyst@elk-server:~$ sudo ufw allow 22/tcp
Rule added
Rule added (v6)
analyst@elk-server:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), deny (routed)
New profiles: skip

To Action From
--
80/tcp ALLOW IN Anywhere
22/tcp ALLOW IN Anywhere
80/tcp (v6) ALLOW IN Anywhere (v6)
22/tcp (v6) ALLOW IN Anywhere (v6)

analyst@elk-server:~$ _
```

7.2 Workstation Endpoint Hardening (Windows 10)

Defensive configurations were executed via PowerShell as Administrator:

- **Inbound ICMP Echo Request Restriction (Ping Blocking):** Enabling Windows Defender Firewall blocks inbound ICMPv4 Echo Requests by default, rendering the workstation silent to standard ICMP ping sweeps (`nmap -sn`).
- **SMB Security Signing Enforcement:** Enforced SMB message signing to mitigate NTLM relay and Man-in-the-Middle (MitM) attacks:

```
Set-SmbServerConfiguration -RequireSecuritySignature $true -Force
```

- **Firewall Inbound Restrictions:** Enforced inbound rules blocking unauthenticated SMB traffic over port 445:

```
New-NetFirewallRule -DisplayName "Block Inbound SMB 445" -Direction Inbound -
Action Block -Protocol TCP -LocalPort 445
```

```
Administrator: Windows PowerShell
PS C:\Windows\system32> Set-SmbServerConfiguration -RequireSecuritySignature $true -Force
PS C:\Windows\system32> Get-WmiObject Win32_NetworkAdapterConfiguration | Where-Object { $_.IPEnabled -eq $true } |
ForEach-Object { $_.SetTcpipNetbios(2) }

__GENUS           : 2
__CLASS           : __PARAMETERS
__SUPERCLASS      :
__DYNASTY         : __PARAMETERS
__RELPATH         :
__PROPERTY_COUNT  : 1
__DERIVATION      : {}
__SERVER          :
__NAMESPACE       :
__PATH            :
ReturnValue       : 0
PSComputerName    :

PS C:\Windows\system32> New-NetFirewallRule -DisplayName "Block Inbound SMB 445" -Direction Inbound -Action Block -
Protocol TCP -LocalPort 445

Name                : {697ea8e8-6190-442d-b3d3-a4aaca4729a9}
DisplayName          : Block Inbound SMB 445
Description          :
DisplayGroup         :
Group                :
Enabled              : True
Profile              : Any
Platform             : {}
Direction            : Inbound
Action               : Block
EdgeTraversalPolicy  : Block
LooseSourceMapping   : False
LocalOnlyMapping     : False
Owner                :
PrimaryStatus        : OK
Status               : The rule was parsed successfully from the store. (65536)
EnforcementStatus    : NotApplicable
PolicyStoreSource    : PersistentStore
PolicyStoreSourceType : Local
RemoteDynamicKeywordAddresses :
PolicyAppId          :
```

8. Post-Hardening Validation Scan

Post-remediation scans were executed from **10.162.238.51** to validate the efficiency of the applied controls across both hosts.

8.1 Verification Commands

```
# Ubuntu Server Post-Hardening Check
nmap -sV -p 21,80,22 10.162.238.10 -oN server_hardened_scan.txt
```

```
> nmap -sV -p 21,80,22 10.162.238.10 -oN server_hardened_scan.txt
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-25 16:39 IST
Nmap scan report for 10.162.238.10
Host is up (0.00037s latency).

PORT      STATE      SERVICE VERSION
21/tcp    filtered  ftp
22/tcp    open      ssh      OpenSSH 10.2p1 Ubuntu 2ubuntu3.5 (Ubuntu Linux; protocol 2.0)
80/tcp    open      http     Apache httpd 2.4.25 ((Debian))
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.43 seconds

~ 7s
>
```

```
# Windows 10 Workstation Post-Hardening Check (Using -Pn due to ICMP block)
nmap -Pn -sV -p 135,139,445,3389 10.162.238.20 -oN win10_hardened_scan.txt
```

```
> nmap -Pn -sV -p 135,139,445,3389 10.162.238.20 -oN win10_hardened_scan_2.txt
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-25 16:53 IST
Nmap scan report for 10.162.238.20
Host is up (0.00038s latency).

PORT      STATE      SERVICE      VERSION
135/tcp    open      msrpc        Microsoft Windows RPC
139/tcp    filtered  netbios-ssn
445/tcp    filtered  microsoft-ds
3389/tcp   filtered  ms-wbt-server
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.38 seconds
```

8.2 Validation Proof

Comparative Findings Matrix

Target Host	Service / Component	Initial State	Post-Hardening State	Remediation Action Taken
Ubuntu Server	TCP 21 (FTP)	OPEN	FILTERED	Service stopped via <code>systemctl stop vsftpd</code> ; UFW drops incoming probes on port 21
Ubuntu Server	TCP 80 (HTTP)	OPEN	OPEN	Retained for legitimate DVWA application traffic
Windows 10	ICMP Echo (Ping)	RESPONSIVE	DROPPED / UNRESPONSIVE	Windows Defender Firewall drops incoming ICMP sweeps
Windows 10	TCP 445 (SMB)	OPEN	FILTERED	Inbound port 445 blocked via Windows Firewall
Windows 10	SMB Signing	Enabled (Not Required)	Required	Mandatory packet signing enforced via PowerShell

Verification confirmed that unencrypted entry vectors were neutralized, ICMP recon sweeps were negated, and vital business applications remained accessible.

9. Hardening Recommendations

- **Enforce Transport Layer Encryption:** Transition all cleartext HTTP web application endpoints to HTTPS (TLS 1.3) using valid SSL/TLS certificates. *The DVWA instance was intentionally deployed as a vulnerable application for security assessment. Its HTTP transport should not be considered suitable for production deployment. In a production environment, authenticated web applications should use HTTPS with appropriately configured TLS.*
- **Deprecate Legacy Protocols:** Permanently purge FTP, Telnet, and unencrypted HTTP from enterprise host templates, enforcing SSH/SFTP exclusively.
- **Network Segmentation:** Place workstation endpoints, servers, and administrative hosts into dedicated Virtual LANs (VLANs) governed by firewall access control lists (ACLs).
- **Centralized Policy Management:** Deploy Active Directory Group Policy Objects (GPO) to enforce SMB signing, firewall profiles, and audit policies domain-wide.

10. Conclusion

This project successfully completed the design, implementation, evaluation, and security hardening of a mini enterprise network. Initial enumeration and packet analysis highlighted critical exposure points associated with cleartext FTP services and unconstrained Windows network protocols, allowing plain text credentials to be harvested over the wire. By implementing UFW stateful firewall policies on Linux, enforcing ICMP stealth and SMB signing on Windows 10, and disabling unneeded daemons, the environment's total attack surface was systematically reduced as verified by post-remediation Nmap scans. This hands-on exercise reinforces fundamental VAPT techniques, packet analysis workflows, and practical defense-in-depth methodologies essential for modern enterprise network security.

11. References

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